

Detecting exoplanets with the radial velocity method

1.1 Introduction

In the Fall of 1995, two Swiss astronomers announced evidence for a planet orbiting the star 51 Pegasi, a groundbreaking discovery since the star, 51 Pegasi, is very similar to our own Sun. Since 1995, thousands of potential exoplanets have been detected and the field of exoplanet science has become a pillar of modern astronomy.

In this lab, we will explore the radial velocity technique, the method used to detect the first exoplanets and a powerful technique for studying exoplanets.

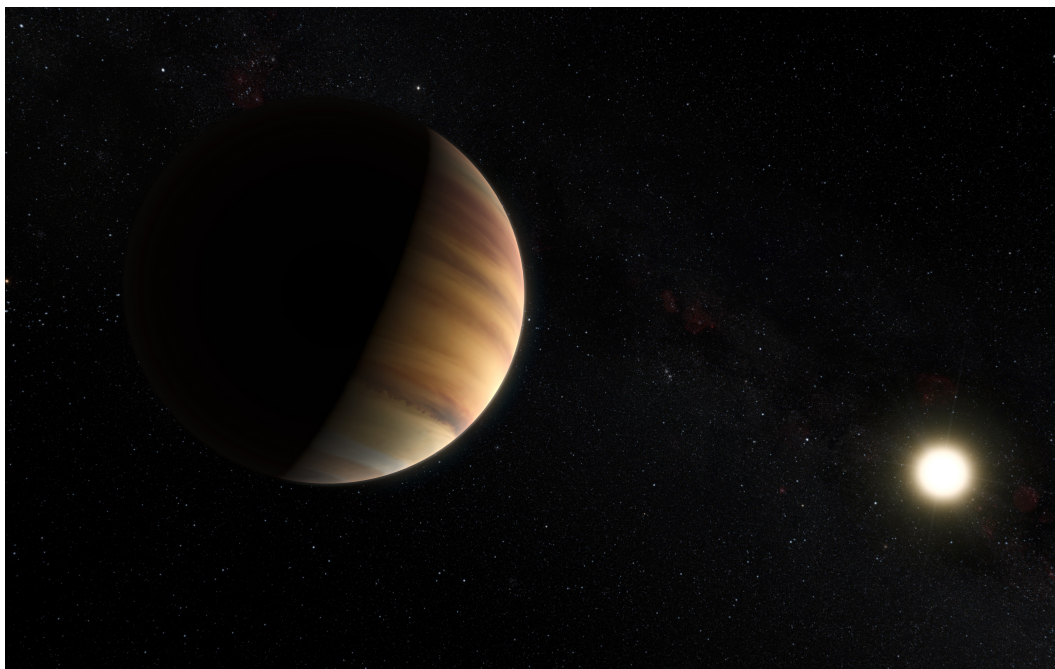


Figure 1.1: Artist impression of the hot Jupiter exoplanet 51 Pegasi b, the first exoplanet found around a normal star. (Source: ESO)

A Note about Collaboration

While each student will be submitting an individual report for labs in this course, you are encouraged to discuss your analysis with peers and collaborate on creating plots and images. **Therefore, your report should include explicit documentation of whether you worked alone or with whom you collaborated.**

At the same time, any written responses should be your own work. See the Lab Syllabus for more details about collaboration within this course.

1.2 Building Intuition

We usually say that a planet orbits a star, but this isn't quite true — both planet and star orbit the center of mass of the system. This is called the *barycenter*. This results in the star wobbling in a circular motion. The motion is too small to detect from Earth directly. But when the source of waves moves towards or away from an observer, the wavelength of that wave is shifted lower or higher, respectively. Since stars have characteristic gaps in the spectrum of light they emit, we can detect this *Doppler shift*, and thus determine the *radial velocity* of the star. From this and our knowledge of orbital dynamics, we can deduce characteristics of exoplanets orbiting that star.

1. Open the link for the [Center of Mass Simulation](#) and use the sliders to explore how varying separation and relative mass changes the Center of Mass. **Include your findings in your report.**

2. Open the link for the [Radial Velocity Simulator](#).

Vary the mass of the planet, the mass of the star, and the distance. Note how the radial velocity graph changes. Vary the other parameters (viewing angle, star radius, planet radius) and see how the radial velocity changes. **Include your findings in your report.**

1.3 Building quantitative intuition

For this lab, the following values will be useful:

- $1 \text{ AU} = 1.496 \times 10^{11} \text{ m}$ (approximate distance between the Earth and the Sun)
- $1 \text{ M}_J = 1.898 \times 10^{27} \text{ kg}$ (mass of Jupiter)
- $1 \text{ M}_{\text{Sun}} = 1.989 \times 10^{30} \text{ kg}$ (mass of the Sun)
- $G = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ (Newtonian constant of gravitation)

Mathematical tools for orbital mechanics

Equation for finding the center of mass position x_{CM} for two masses, m_1 and m_2 , at positions x_1 and x_2 :

$$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \quad (1.1)$$

An object in a circular orbit of radius r experiences an acceleration $\vec{a}$ directed inward with magnitude

$$a = \frac{v^2}{r}. \quad (1.2)$$

In our case, the object is experiencing this acceleration as a result of the force exerted on it through gravity by the other object according to

$$F = G \frac{m_1 m_2}{r^2}, \quad (1.3)$$

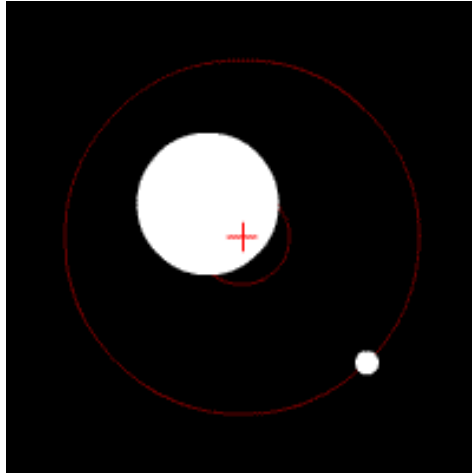


Figure 1.2: Schematic of a star (upper-left) and planet (lower-right) orbiting a common center-of-mass (marked with a red cross).

where r_{12} is the distance between the two objects, which is different from the orbital radius, since the object orbits the center of mass (or barycenter), rather than the other object.

To relate velocity in Equation 1.2 with the masses of the objects in Equation 1.3, we can use the fact that the acceleration that an object experiences is proportional to the force on it and inversely proportional to its mass — or in equation form,

$$a = \frac{F}{m}, \quad (1.4)$$

also known as Newton's second law of motion.

So in this equation, if we substitute in a from Eq. 1.2 and F from Eq. 1.3, we can solve for the orbital velocity of Object 1 like so:

$$\frac{v^2}{r} = \frac{G \frac{m_1 m_2}{r_{12}^2}}{m_1} \quad (1.5)$$

$$\frac{v^2}{r} = G \frac{m_2}{r_{12}^2} \quad (1.6)$$

Finally, if we assume an orbit is roughly circular, we can related the orbital velocity v , radius r , and period T using the equation:

$$v = \frac{2\pi r}{T} \quad (1.7)$$

Analyzing a sample orbital system

Consider the picture in Figure 1.2 and define $M_p = 0.5M_J$ as the mass of the planet (the smaller object), $M_S = 1.0M_{\text{Sun}}$ as the mass of the star (the larger object), and $r_{12} = 0.0527$ AU be the separation between the two. Assume the system is in a circular orbit.

3. How far is the Center of Mass from the center of the star (in AU)?
4. What is the star's orbital radius, r (in AU)?

Since the star is moving around the Center of Mass, it has a non-zero velocity. Let the orbital period of the system be $P = 4.42$ days.

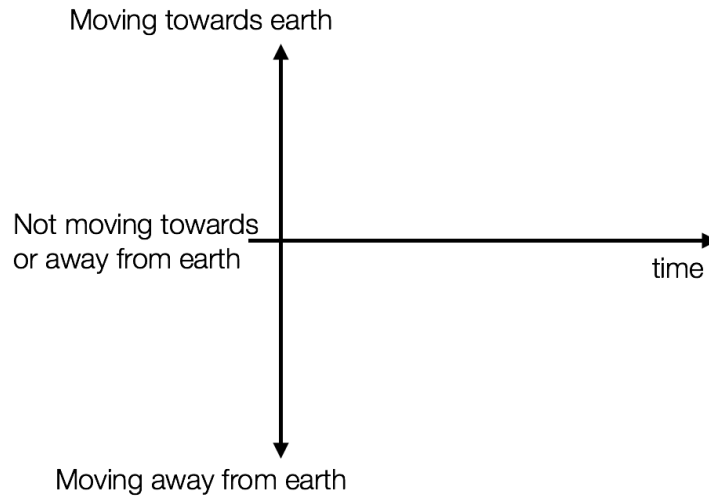


Figure 1.3: Graph to draw for radial velocity estimation.

5. What is the orbital velocity of the host star?
6. Sketch the figure, then on that sketch, draw arrows showing the velocity at different parts of the orbit. What happens to the direction of the star's velocity as the planet traces out its orbit?

Check your answers to the previous 4 questions with your TA to confirm you are correct before moving on.

Astronomers can use spectroscopic techniques for measuring the velocity of stars along the line of sight between the earth and the star. In the illustration in Figure 1.2, suppose the earth is to the right of the system so that we are observing the system edge on from the right side of the page.

7. Draw a graph similar to Figure 1.3 and then draw on it the “line of sight” velocity for the star as a function of time (hint: it will be a trigonometric function).
8. How are the period and amplitude related to the orbital period and the star's velocity?

Next, consider a similar system only with a planet that is twice the mass ($M_p = M_J$, $M_S = M_{\text{Sun}}$, $d = 0.0527$ AU, and $P = 4.42$ days).

9. What is the system's host star orbital radius (in AU) and orbital velocity (in m/s)?
10. How do these compare with the values for the previous system with a less massive planet?
11. Sketch the line of sight velocity for this more massive system on the same plot that you already drew.
12. How does the line of sight velocity depend on the planet's mass?

Check your answers to the previous 6 questions with your TA to confirm you are correct before moving on.

DAY	v (m/s)	DAY	v (m/s)	DAY	v (m/s)	DAY	v (m/s)	DAY	v (m/s)	DAY	v (m/s)	DAY	v (m/s)
0.6	-20.2	3.6	-35.1	5.6	45.3	7.7	-22.6	9.6	25.1	10.8	51	13.6	2.7
0.7	-8.1	3.7	-42.6	5.7	47.6	7.8	-31.7	9.7	35.7	11.7	-2.5	13.7	17.6
0.8	5.6	4.6	-33.5	5.8	56.2	8.6	-44.1	9.8	41.2	11.8	-4.6		
1.6	56.4	4.7	-27.5	6.6	65.3	8.7	-37.1	10.6	61.3	12.6	-38.5		
1.7	66.8	4.8	-22.7	6.7	62.5	8.8	-35.3	10.7	56.9	12.7	-48.7		

Table 1.1: Line-of-sight velocities for 51 Pegasi measured over time.

1.4 Radial velocity of 51 Peg

Table 1.1 lists measured line of sight velocities for the star 51 Pegasi measured by Marcy & Butler in 1995 at the Lick Observatory in California. You will use this data to determine the mass and orbital distance of the exoplanet orbiting 51 Peg.

- Open this file of [51 Pegasi data](#). Make a copy of this sheet for your group to use.

This spreadsheet contains the data in Table 1.1, as well as a model for the velocity as a function of time. Circular motion, projected to 1 dimension, can be described by a sine function. The most general sine function has free parameters that can be changed to represent any sinusoidal motion. For example, the velocity function can be described with

$$v(t) = A \sin \left[\frac{2\pi}{B} (t + C) \right] + D, \quad (1.8)$$

where A is the amplitude of the motion and B is the period of the motion. C is a time offset, to account for where the star is in its oscillation cycle at $t = 0$. D is constant velocity offset, to account for any constant line of sight velocity that was not already compensated for (for example, the observer's motion with respect to the barycenter of the exoplanet system).

- In your copy of the spreadsheet, explore changing the fit parameters A – D and see what effect it has on the model velocity.
- For your first pass at fitting the model, find a set of parameters that makes the model velocity curve visually similar to the data.

To get a more precise estimate for the parameters, you can use a common statistic called the *rms deviation*. "rms" stands for root-mean-square. For N data points $y_{\text{data},i}$, the rms deviation from the theoretical model points $y_{\text{model},i}$ can be found by

$$\text{rms deviation} = \sqrt{\frac{\sum_{i=0}^{N-1} (y_{\text{data},i} - y_{\text{model},i})^2}{N}}. \quad (1.9)$$

The smaller the rms deviation, the closer the fit. The rms deviation is already calculated for you in cell D6 in the spreadsheet.

- Starting with your first pass estimate of parameters, vary them to minimize the rms deviation and find your best fit parameters. **Record these parameters, the resulting plot, and your final rms deviation.**

17. From your fit to the data, what is the orbital period (days) and host star orbital velocity (m/s) – assuming the planetary orbit is edge-on so the radial velocity amplitude equals its full velocity?
18. What is the orbital radius (in AU) for the host star, 51Peg? Justify your answer with a calculation or explanation.

Kepler’s third law relates the orbital period to the system’s radius¹. In the case where the planet’s mass is much smaller than the star’s mass, Kepler’s third law is

$$T^2 = \frac{4\pi^2}{GM} r^3, \quad (1.10)$$

where T is the orbital period, G is Newton’s constant, M is the mass of the star, and r is the orbital radius. The mass of 51Peg is $1.1M_{\text{Sun}}$.

19. According to your fitted orbital period and Kepler’s third law, what is the planet’s orbital radius?
20. Using Equation 1.1 with the planet’s orbital radius and 51Peg’s orbital radius & mass, determine the planet’s mass (in units of M_J). Of course, we really mean its minimum mass, $m \sin i$, due to the assumption of an edge-on orbit.

1.5 Radial velocities for other systems

21. Go to the webpage for [SystemicLive](#).
22. Click “Open Systemic” to start the program. Then on that first page, go to the drop down menu in the upper left hand corner labeled “System” and select “51peg.sys”
23. In the upper right hand corner of the same screen, select the drop down menu at the far upper right (labeled *Systemic Live*) and select the “Tutorials” option. This will open another tab with the title “51 Pegged: Rediscovering the first exoplanet with Systemic Live”. keep both tabs open.
24. Follow the steps of the 51 Pegged tutorial to analyze radial velocity data for 51Peg in your other tab. Note that while the graphics within the tutorial are currently not visible on the website, the data visualization and analysis software are still functional.

Note that one key analysis tool you will use in the remainder of the lab is the power spectrum of the radial velocity data. Any periodic function from the sum of sinusoids of different frequencies and phases: that’s why Ptolemy’s mechanics of epicycles could fit the motion of the planets so well! The power spectrum tells us which frequencies *dominate* that decomposition. Therefore, peaks in the power spectrum correspond to periodicities in the data, and hence point to possible planetary periods in the radial velocity measurements, with smaller peaks being attributable to noise.

25. How do the mass and separation results from Systemic compare with your earlier 51Peg numbers? Screenshot a plot of the “Phased Radial Velocity” and notate on it the Period and Mass for the system.
26. Use Systemic to determine orbit parameters of one of the planets (there are multiple present) in each of the following systems: 47Uma, 70Vir. Screenshot plots for your “Phased Radial Velocity” for each system and notate the Period and Mass for each system orbit.

¹Since all orbits are ellipses rather than circles (according to Kepler’s 1st Law), this should technically be semi-major-axis. However, the math is much simpler and the principle largely unchanged if we model the orbit as circular with a set radius.

27. Use **Systemic** to analyze the data for the **Upsilon Andromedae** system (noted as “upsand”). How many planets do you find? What are their masses and orbital periods?

1.6 Report checklist

Include the following in your lab report. See Appendix B for formatting details. Each item below is worth 10 points.

1. Discussion of how radial velocity changes based on available parameters (Steps 1–2)
2. Your plots for line-of-sight velocities and fit parameters (Step 16)
3. The host star’s orbital period, velocity, and orbital radius with justification for the radius. (Steps 17–18)
4. The exoplanet’s orbital radius and mass with justification, as well as comparison to Systemic analysis (Steps 19, 20, & 25)
5. Orbit parameters for 3 star-planet systems (Steps 26–27)